

Sampling, FFT, and the Pre-Lab 1 notebook

Fall 2026 · Prof. Ehsan Roohi

Lecture and guided Jupyter walkthrough

Learning goals

- Explain sample rate, sample interval, record length, and Nyquist frequency
- Run prepared analysis cells in a Jupyter notebook
- Interpret time histories, RMS values, and one-sided FFT spectra
- Recognize aliasing before collecting Lab 1 sound data

Connection to Lab 1

PRE-LAB

A known voltage signal contains two pure tones.

Amplitude describes signal size; frequency describes how fast it repeats; phase sets its starting position.

LABORATORY

A microphone converts changing sound pressure into changing voltage.

Moku:Go stores voltage samples. The waveform shows change in time; the spectrum shows the frequencies present.

Jupyter notebook structure

MARKDOWN CELLS

Instructions, equations, and response areas.

Double-click to edit. Use Shift+Enter to render and move forward.

CODE CELLS

Python commands that define variables, calculate results, and make figures.

Run from top to bottom with Shift+Enter.

Opening and saving the notebook

- Download the .ipynb file from Canvas or open it from the course website in Colab
- Save a personal copy before editing
- Run cells from top to bottom
- Keep code outputs and figures in the submitted file

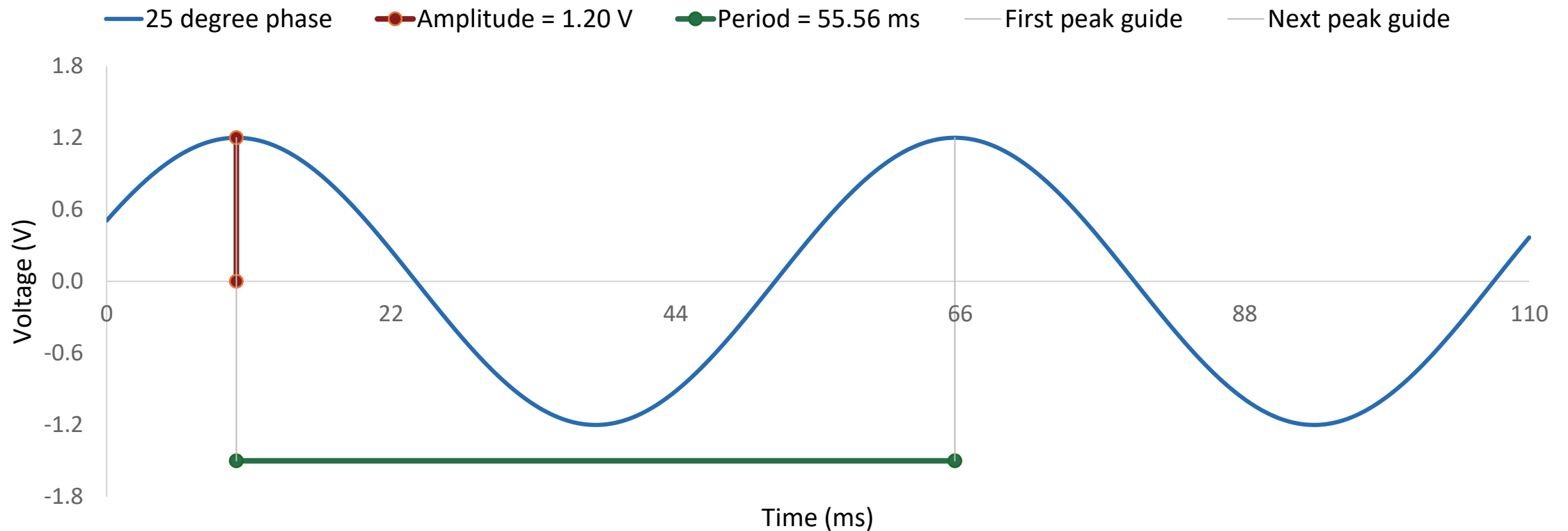
Execution order matters

```
import numpy as np
A = 1.20
print(A)

# A later cell can use A only after this cell runs.
rms_component = A / np.sqrt(2)
print(rms_component)
```

A restart clears stored variables. Run All rebuilds the notebook state from the first cell.

A sinusoid: what the three parameters mean



$A_1 = 1.20$ V from zero to peak. $T_1 = 55.56$ ms between consecutive peaks.

$$x_1(t) = A_1 \sin(2\pi f_1 t + \phi_1), \text{ with } f_1 = 18 \text{ Hz and } \phi_1 = 25^\circ$$

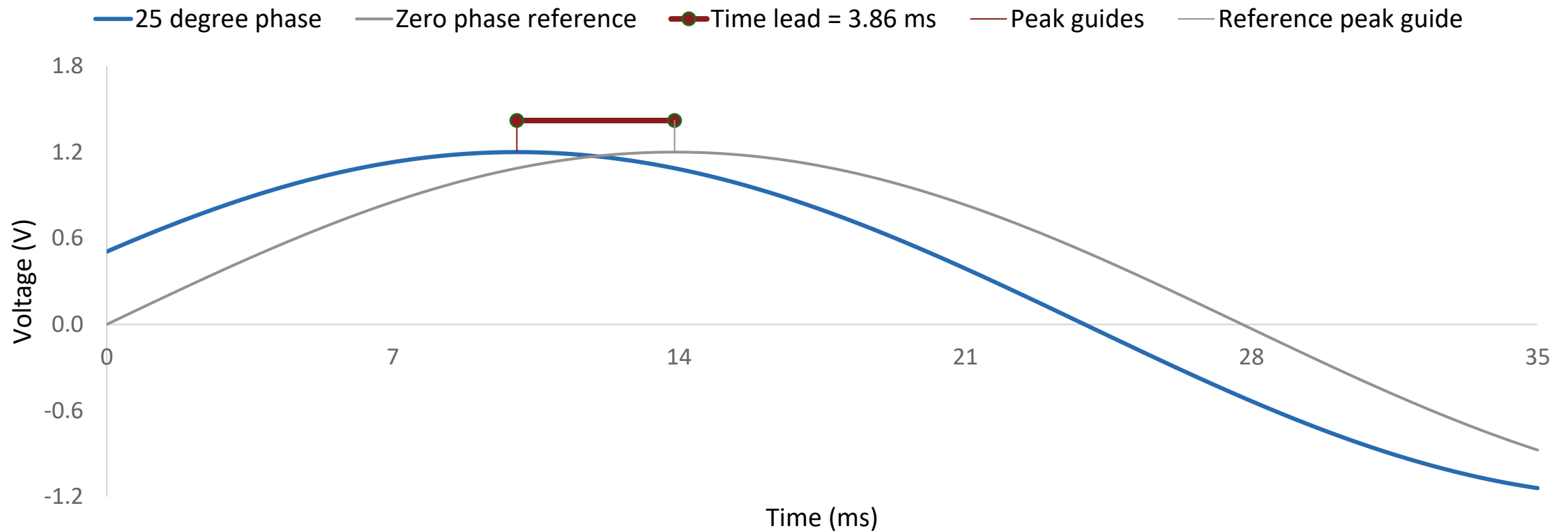
The new two-tone signal

```
A1, f1, phi1_deg = 1.20, 18.0, 25.0
A2, f2, phi2_deg = 0.45, 42.0, -15.0

def signal(t):
    return (A1*np.sin(2*np.pi*f1*t + np.deg2rad(phi1_deg))
            + A2*np.cos(2*np.pi*f2*t + np.deg2rad(phi2_deg)))
```

Subscript 1 means tone 1; subscript 2 means tone 2. A is in volts, f is in hertz, and phi_deg is in degrees.

Phase relative to a zero-phase sine wave



$\phi_1 = +25^\circ$: the blue wave reaches its peak 3.86 ms earlier

Time lead = $\phi_1 / (360^\circ f_1)$. Positive phase shifts a sine wave to the left.

Mean and RMS

MEAN

Average signed level:

$$\text{mean} = (1/N) \sum x[n]$$

N is the sample count; $x[n]$ is sample n.

The continuous one-second signal has zero mean.

RMS

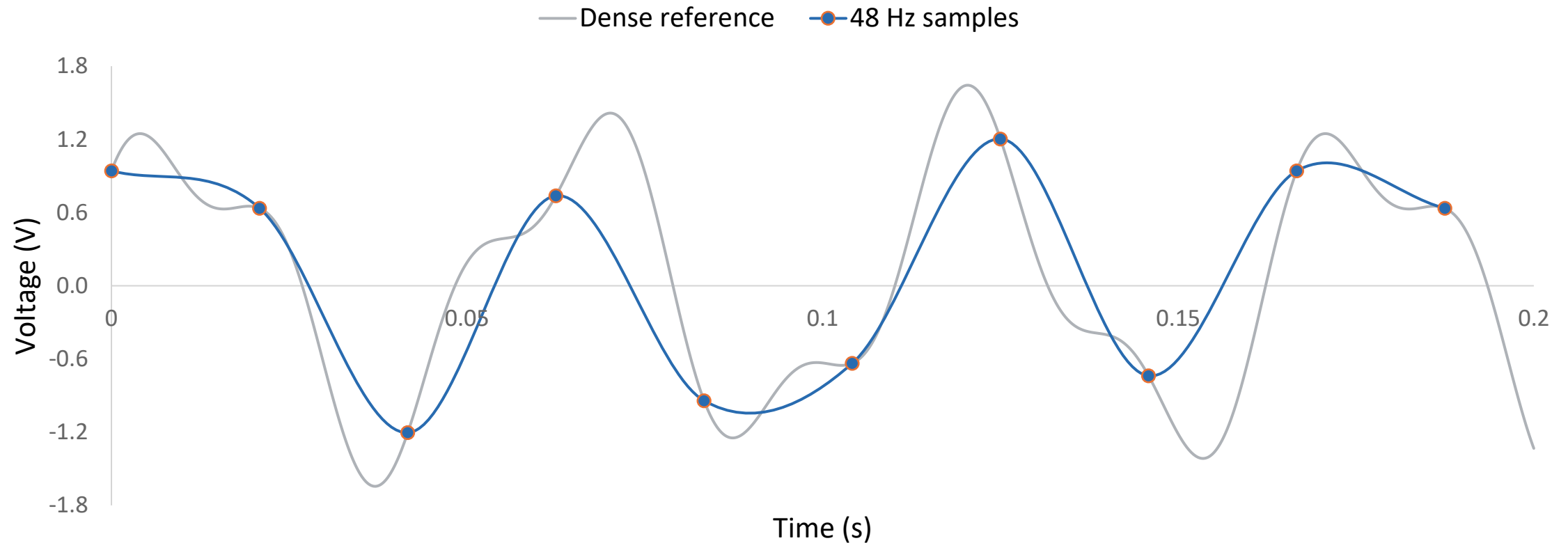
Typical signal size, independent of sign:

$$\text{RMS} = \sqrt{(1/N) \sum x[n]^2}$$

For these two distinct tones over complete cycles:

$$\text{RMS} = \sqrt{(A_1^2 + A_2^2)/2} = 0.9062 \text{ V}$$

Samples and the underlying waveform



Markers are simulated samples of the known signal. Connecting lines add no measurements.

Sample rate, interval, and record length

```
fs = 126.0                # samples per second
dt = 1/fs                 # seconds between samples
duration = 1.00           # seconds
n = int(round(duration*fs))
t = np.arange(n)/fs

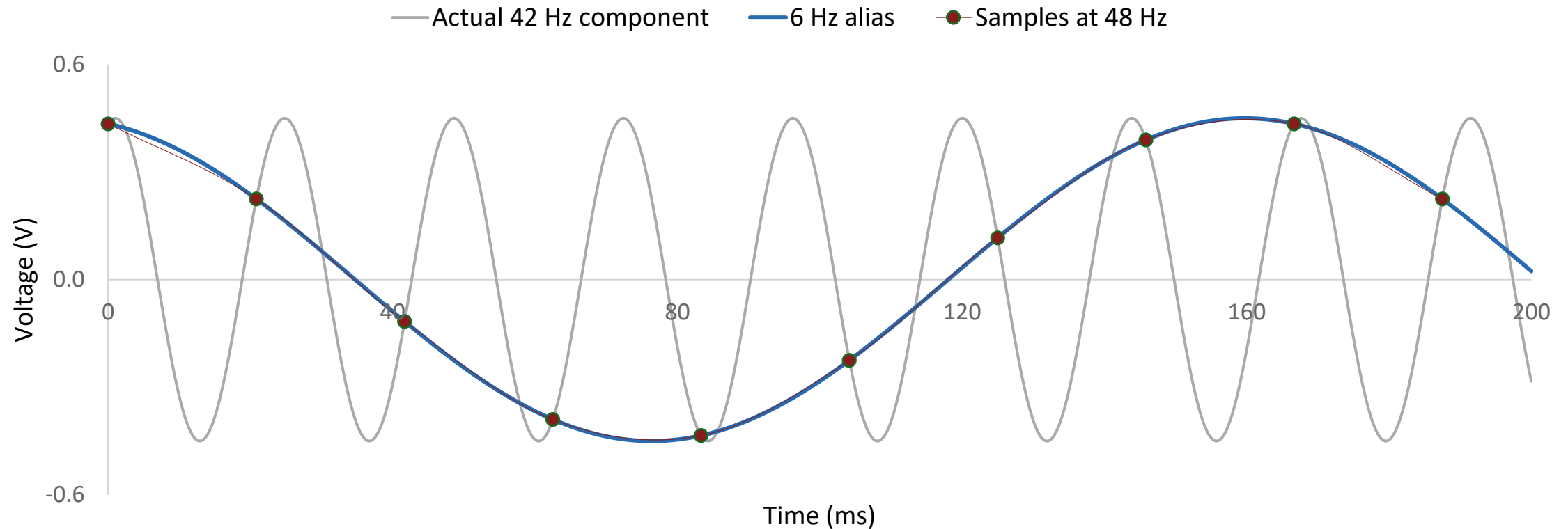
print(dt, n, t[-1])
```

$\Delta t = 7.94 \text{ ms}$; $N = 126$; last sample = 0.9921 s. FFT record duration = $N/fs = 1.00 \text{ s}$.

Nyquist frequency

- Samples per cycle = f_s / f_{signal}
- For a baseband signal, $f_N = f_s/2$ is the highest unaliased frequency boundary
- For the 42 Hz tone, f_s must be greater than 84 Hz
- 336 and 126 Hz satisfy $f_s > 2 f_{\text{max}}$. Real acquisition also needs anti-alias filtering

Aliasing changes the measured frequency



Both waves give the same sample values at $f_s = 48$ Hz

42 Hz folds to $|42 - 48| = 6$ Hz. The signal itself has not changed.

The prepared sampling analysis

```
for fs in sample_rates:
    n = int(round(duration*fs))
    t = np.arange(n)/fs
    x = signal(t)
    records[fs] = (t, x)
    sample_mean, sample_rms = mean_and_rms(x)
    print(fs, sample_mean, sample_rms)
```

Students transfer the printed values to an analysis table and calculate RMS percent error.

Reading the time-domain figures

- Compare sample markers with the dense reference over the same time window
- Look for too few points per cycle and apparent changes in shape
- Read the printed mean and RMS values
- Avoid judging sample quality from smooth connecting lines

FFT peaks, windowing, and amplitude

READING THE SPECTRUM

Horizontal axis: frequency (Hz).

Peak location identifies a tone.

Peak height estimates amplitude (V).

One-sided means nonnegative frequencies for a real-valued signal.

WHY USE A WINDOW?

A finite record may end mid-cycle.

This spreads a tone across FFT bins (leakage).

A Hann window reduces leakage but widens peaks. Gain correction adjusts amplitude scaling.

Record duration and FFT-bin spacing

SHORT RECORD: $T = 0.25$ s

Bin spacing $\Delta f = 1/T = 4$ Hz

For $N = 84$ at $f_s = 336$ Hz.

18 Hz and 42 Hz lie between bins, so their peaks spread.

LONGER RECORD: $T = 1.00$ s

Bin spacing $\Delta f = 1/T = 1$ Hz

For $N = 336$ at $f_s = 336$ Hz.

18 Hz and 42 Hz fall on bins. Resolution also depends on the window.

Expected FFT peaks

ADEQUATE SAMPLE RATE

336 Hz gives peaks at 18 and 42 Hz.

126 Hz gives peaks at 18 and 42 Hz.

ALIASING

72 Hz gives peaks at 18 and 30 Hz.

48 Hz gives peaks at 18 and 6 Hz.

Guided notebook walkthrough

- 1 Define the signal**
- 2 Choose f_s and create sample times**
- 3 Plot samples beside the reference**
- 4 Compute the one-sided FFT**
- 5 Explain the physical meaning**

Required data analysis

- Calculate theoretical mean and RMS, then compute RMS percent errors
- Predict every FFT peak before viewing the spectra
- Compare predicted and observed peaks and explain aliasing
- Choose a sample rate and duration for a new measurement and justify both

Common notebook problems

CODE PROBLEMS

NameError: run earlier cells.

Missing figure: run the plotting cell.

Changed variable: restart and Run All.

ANALYSIS PROBLEMS

Wrong units for phase.

Confusing f_s with signal frequency.

Calling a smooth line measured data.

Reporting peaks without explaining aliasing.

From the notebook to Moku:Go

- Choose a sample rate from the known or estimated sound frequency
- Choose record duration for the required frequency detail, then export to MAT
- Plot the measured waveform and compute its spectrum
- Compare peaks with the source frequency. Replay audio at its recorded sample rate when supported

Pre-Lab 1 submission

BEFORE SUBMITTING

Run All cells.

Keep figures visible.

Complete every response section.

Export to HTML or PDF for Canvas.

DEADLINE

Normally posted Monday before the laboratory.

Submit before your own section begins.

Late pre-labs are not accepted.